程式實作方面，TODO 1 完成三個尺度的偵測 head，透過卷積、上採樣與特徵串接，融合不同解析度的資訊，分別在 13×13、26×26、52×52 網格預測不同大小的物件。每個位置包含三個 anchor，各輸出框的位置、物件存在分數與類別分數。

Loss 部分，TODO 2 實作可選用的 MSE 定位損失；TODO 3 完成實驗實際使用的 GIoU loss，透過交集、聯集與最小包圍矩形衡量預測框與標註框的差異。TODO 4 將各尺度的定位、物件存在與分類損失累加，其中定位與分類只計算有物件的位置，背景則只計算物件存在損失，並分開正規化，避免大量背景主導訓練。

實驗保留 YOLO 架構，先使用 darknet53.c2ns_in1k 訓練 50 epochs，再改用 convnext_large.fb_in1k，將訓練設定增加至 80 epochs。ConvNeXt-Large 組使用 batch size 8、learning rate 0.0001，搭配 AdamW、cosine learning-rate schedule 與混合精度訓練。

Darknet53 的 AP/AP50 為 0.1209/0.2515，ConvNeXt-Large 則為/0.1672/0.3312，分別提升 4.63 與 7.97 個百分點。Notebook 已記錄的最高 AP 為第 60 個 epoch 的 0.1723。不過，兩組同時改變 backbone 與訓練輪數，因此提升不能完全歸因於模型大小。

訓練後期雖然 training loss 持續下降，validation loss 卻逐漸上升，顯示過擬合跡象；小物件 AP 也明顯低於大型物件。後續可加強資料增強、調整輸入解析度，並額外依 validation AP 儲存最佳模型，避免最低 loss 的權重未必具有最佳偵測表現。

若進一步更換架構，可嘗試 DINO 搭配 ImageNet-1K 預訓練的 ResNet backbone，其餘偵測部分使用作業資料從頭訓練，可能改善特徵互動與預測配對效果。但此方向尚未實驗，仍需以相同條件比較，才能確認是否優於目前的 YOLO。

convnext_large.fb_in1k 訓練 80 epoch

```
creating index...
index created!
Loading and preparing results...
DONE (t=1.12s)
creating index...
index created!
Running per image evaluation...
Evaluate annotation type *bbox*
DONE (t=16.77s).
Accumulating evaluation results...
DONE (t=4.61s).
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.167
 Average Precision  (AP) @[ IoU=0.50      | area=   all | maxDets=100 ] = 0.331
 Average Precision  (AP) @[ IoU=0.75      | area=   all | maxDets=100 ] = 0.154
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.051
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.198
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.347
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=  1 ] = 0.217
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets= 10 ] = 0.330
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.349
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.172
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.382
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.542
Validation AP: 0.1672 | AP50: 0.3312
```

darknet53.c2ns_in1k 訓練 50 epoch

```
creating index...
index created!
Loading and preparing results...
DONE (t=1.15s)
creating index...
index created!
Running per image evaluation...
Evaluate annotation type *bbox*
DONE (t=16.13s).
Accumulating evaluation results...
DONE (t=4.77s).
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.121
 Average Precision  (AP) @[ IoU=0.50      | area=   all | maxDets=100 ] = 0.251
 Average Precision  (AP) @[ IoU=0.75      | area=   all | maxDets=100 ] = 0.104
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.037
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.134
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.250
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=  1 ] = 0.175
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets= 10 ] = 0.283
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.301
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.142
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.329
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.459
Validation AP: 0.1209 | AP50: 0.2515
```